

SOLANDER ISLAND, BC, CanadaWMO: **714790**Lat: **50.111N**Lon: **127.940W**Elev: **307**StdP: **14.53**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.3	33.7	15.5	12.2	34.6	21.1	16.0	38.8	73.7	42.3	66.7	43.2	19.1	10	1.018

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	63.3	57.5	61.0	57.2	59.6	56.7	59.7	61.9	58.6	60.2	57.8	59.3	11.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.9	75.2	60.1	57.9	72.5	59.1	57.0	70.2	58.3	26.3	62.2	25.6	60.4	25.0	59.4	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 64.4	56.6	49.1	28.0	72.8	5.4	6.0	24.1	77.1	20.9	80.7	17.9	84.0	14.0	88.4		
(5)			24.1	62.1	5.2	2.1	20.4	63.6	17.3	64.8	14.4	66.0	10.6	67.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	43.2	42.2	42.7	45.2	48.5	51.6	54.2	55.8	55.3	50.5	46.2	42.9
	DBStd	6.17	4.17	3.97	3.99	3.37	3.49	3.62	3.62	3.54	3.07	3.18	4.18	4.02
	HDD50	1297	213	220	228	149	68	21	12	7	2	31	124	222
	HDD65	6130	677	640	690	593	511	404	335	287	292	450	565	686
	CDD50	644	1	1	3	6	22	68	142	185	161	45	9	1
	CDD65	3	0	0	0	0	0	1	0	1	1	0	0	0
	CDH74	13	0	0	0	0	0	3	5	3	2	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
	WSAvg	22.1	26.3	23.2	23.0	21.1	20.1	18.8	20.4	19.7	19.3	22.4	24.7	25.9
(15) Precipitation	PrecAvg	107.2	13.5	9.2	10.4	7.3	4.5	3.5	3.0	4.2	6.3	14.0	16.0	15.4
	PrecMax	139.6	23.6	22.6	18.0	16.3	7.4	6.0	8.3	9.5	12.5	25.2	23.5	27.2
	PrecMin	85.7	6.4	3.1	3.0	2.1	1.4	1.3	0.7	0.9	1.5	4.0	8.8	6.6
	PrecStd	12.6	3.9	4.1	3.2	2.7	1.5	1.2	1.8	2.2	2.8	5.2	4.8	4.1
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	53.8	57.2	60.7	62.9	67.3	64.7	68.5	67.5	60.3	56.7	52.6
		MCWB	47.1	45.8	45.9	49.2	53.1	55.2	58.9	58.9	58.8	53.5	51.4	47.9
	2%	DB	50.4	50.1	51.1	54.0	56.6	59.0	60.9	62.5	62.2	57.6	54.1	50.7
		MCWB	47.3	46.0	46.6	48.7	51.4	55.0	57.4	58.6	57.6	54.1	51.8	47.7
	5%	DB	49.2	48.3	49.1	50.7	53.9	56.6	59.4	60.7	60.1	55.8	52.7	49.2
		MCWB	46.7	45.5	46.4	47.3	50.4	53.9	56.8	58.1	56.9	53.4	50.5	47.0
	10%	DB	48.0	47.1	47.6	49.0	52.3	55.2	58.1	59.4	58.9	54.6	51.3	48.0
		MCWB	46.1	44.8	45.3	46.3	49.8	53.1	56.2	57.4	56.2	52.6	49.1	46.1
	0.4%	WB	50.3	50.0	50.2	52.3	54.5	57.9	60.2	61.2	61.1	57.9	55.0	51.1
		MCDB	51.2	50.9	50.6	55.8	59.9	61.7	64.6	63.2	64.8	58.4	55.3	51.5
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.7	48.1	48.6	50.1	52.5	56.2	58.7	59.9	59.2	55.8	52.7	49.3
		MCDB	49.4	49.0	49.9	53.0	55.5	58.6	60.3	61.8	61.4	56.3	53.1	49.9
	5%	WB	47.7	46.7	47.2	48.5	51.3	54.9	57.9	58.9	57.8	54.5	51.1	47.9
		MCDB	48.5	47.6	48.9	50.2	53.6	56.7	59.4	60.5	59.3	55.1	51.7	48.8
	10%	WB	46.9	45.4	45.9	47.1	50.4	53.9	57.0	58.1	56.9	53.3	49.7	46.6
		MCDB	47.7	46.5	47.6	48.7	52.3	55.6	58.3	59.6	58.1	54.2	50.5	47.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	5.7	5.4	4.7	4.5	4.3	4.7	4.9	4.9	5.1	5.1
		MCDBR	5.9	6.6	7.0	7.8	7.6	6.9	6.6	7.1	7.2	6.3	5.8	5.9
		MCWBR	4.2	4.5	5.1	5.3	5.1	4.7	4.5	4.9	5.4	5.3	5.8	5.6
	5% WB	MCDBR	4.9	5.1	5.9	7.1	6.6	6.1	6.4	6.9	6.6	5.4	5.5	5.8
		MCWBR	4.5	4.4	4.9	5.2	4.8	4.4	4.6	4.9	5.3	5.5	6.1	6.0
(40) Clear-Sky Solar Irradiance	taub		0.321	0.325	0.345	0.375	0.380	0.366	0.359	0.362	0.353	0.342	0.327	0.316
	taud		2.444	2.505	2.460	2.360	2.385	2.461	2.526	2.526	2.534	2.508	2.431	2.397
	Ebn at Noon		228	261	274	276	279	283	283	278	268	250	222	210
	Edh at Noon		20	23	29	36	36	34	31	30	27	23	20	18
(44) All-Sky Solar Radiation	RadAvg		246	526	832	1285	1611	1715	1734	1518	1097	601	299	202
	RadStd		30	80	101	96	151	169	179	134	89	79	46	27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-522	+191	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page